

DW01_ash

Arithmetic Shifter

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Parameterized shift coefficient width
- Inferable using a function call

Revision History

Description

DW01_ash is a general-purpose arithmetic shifter. The input data **A** is shifted left or right by the number of bits specified by the control input **SH**.

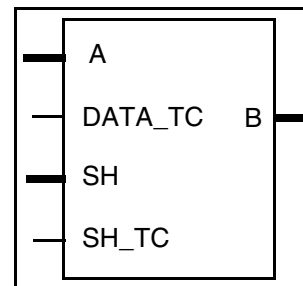


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i> bits	Input	Input data
DATA_TC	1 bit	Input	Data two's complement control ■ 0 = Unsigned ■ 1 = Signed
SH	<i>SH_width</i> bits	Input	Shift control
SH_TC	1 bit	Input	Shift two's complement control ■ 0 = Unsigned ■ 1 = Signed
B	<i>A_width</i> bits	Output	Output data

Table 1-2 Parameter Description

Parameter	Values	Description
A_width	≥ 2	Word length of A and B
SH_width	≥ 1	Word length of SH

Table 1-3 Synthesis Implementations^a

Implementation	Function	License Feature Required
mx2	Implement using 2:1 multiplexers only.	none
str	Synthesis model	DesignWare
astr	Synthesis model	DesignWare

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW01.DW01_ASH_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_ash_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_ash.v	Verilog simulation model source code

When the control signal $SH_TC = 0$, the coefficient SH is interpreted as an unsigned positive number and DW01_ash performs only left shift operations.

When $SH_TC = 1$, SH is a two's complement number, with a negative coefficient performing a right shift and a positive coefficient performing a left shift.

The input data A is interpreted as an unsigned number when $DATA_TC = 0$ or a two's complement number when $DATA_TC = 1$.

The type of A is only significant for right shift operations, where zero padding is done on the most significant bits for unsigned data and sign extension is done for signed two's complement data.

Table 1-5 Truth Table (*SH_width* = 3, *A_width* = 8)

SH(2:0)	SH_TC	DATA_TC	B(7)	B(6)	B(5)	B(4)	B(3)	B(2)	B(1)	B(0)
000	0	X	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)
001	0	X	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	0
010	0	X	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	0	0
011	0	X	A(4)	A(3)	A(2)	A(1)	A(0)	0	0	0
100	0	X	A(3)	A(2)	A(1)	A(0)	0	0	0	0
101	0	X	A(2)	A(1)	A(0)	0	0	0	0	0
110	0	X	A(1)	A(0)	0	0	0	0	0	0
111	0	X	A(0)	0	0	0	0	0	0	0
000	1	X	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)
001	1	X	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	0
010	1	X	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	0	0
011	1	X	A(4)	A(3)	A(2)	A(1)	A(0)	0	0	0
100	1	0	0	0	0	0	A(7)	A(6)	A(5)	A(4)
101	1	0	0	0	0	A(7)	A(6)	A(5)	A(4)	A(3)
110	1	0	0	0	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)
111	1	0	0	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)
100	1	1	A(7)	A(7)	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)
101	1	1	A(7)	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)
110	1	1	A(7)	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)
111	1	1	A(7)	A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DW_foundation_arith.all;

entity DW01_ash_func is
  generic(func_A_width:integer:=8; func_SH_width : integer := 3);
  port(func_A: in std_logic_vector(func_A_width-1 downto 0);
        func_DATA_TC : in std_logic;
        func_SH: in std_logic_vector(func_SH_width-1 downto 0);
        func_SH_TC : in std_logic;
        B_func: out std_logic_vector(func_A_width-1 downto 0));
end DW01_ash_func;

architecture func of DW01_ash_func is
begin
  process (func_DATA_TC, func_SH_TC, func_A,func_SH)
  begin
    if func_DATA_TC = '0' and func_SH_TC = '0' then
      B_func <= std_logic_vector(DWF_ash(unsigned(func_A),unsigned(func_SH)));
    elsif func_DATA_TC = '1' and func_SH_TC = '0' then
      B_func <= std_logic_vector(DWF_ash(signed(func_A),unsigned(func_SH)));
    elsif func_DATA_TC = '1' and func_SH_TC = '1' then
      B_func <= std_logic_vector(DWF_ash(signed(func_A),signed(func_SH)));
    else
      B_func <= std_logic_vector(DWF_ash(unsigned(func_A),signed(func_SH)));
    end if;
  end process;
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW01_ash_func (func_A, func_DATA_TC, func_SH, func_SH_TC, B_func);
    parameter func_A_width = 8;
    parameter func_SH_width = 3;

    // Passes the widths to the arithmetic shifter function
    parameter A_width = func_A_width;
    parameter SH_width = func_SH_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW01_ash_function.inc"

    input [func_A_width-1:0] func_A;
    input [func_SH_width-1:0] func_SH;
    input func_DATA_TC, func_SH_TC;
    output [func_A_width-1:0] B_func;
    reg [func_A_width-1:0] B_func;

    // infer DW01_ash
    always @ (func_A or func_DATA_TC or func_SH or func_SH_TC)
    begin
        casex({func_DATA_TC,func_SH_TC}) // synopsys full_case
            2'b00: B_func = DWF_ash_uns_uns(func_A,func_SH);
            2'b10: B_func = DWF_ash_tc_uns(func_A,func_SH);
            2'b01: B_func = DWF_ash_uns_tc(func_A,func_SH);
            2'b11: B_func = DWF_ash_tc_tc(func_A,func_SH);
        endcase
    end
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_ash_inst is
  generic ( inst_A_width : POSITIVE := 8;
            inst_SH_width : POSITIVE := 8 );
  port ( inst_A      : in std_logic_vector(inst_A_width-1 downto 0);
        inst_DATA_TC : in std_logic;
        inst_SH      : in std_logic_vector(inst_SH_width-1 downto 0);
        inst_SH_TC   : in std_logic;
        B_inst       : out std_logic_vector(inst_A_width-1 downto 0) );
end DW01_ash_inst;

architecture inst of DW01_ash_inst is
begin

  -- Instance of DW01_ash
  U1 : DW01_ash
    generic map ( A_width => inst_A_width, SH_width => inst_SH_width )
    port map ( A => inst_A, DATA_TC => inst_DATA_TC, SH => inst_SH,
              SH_TC => inst_SH_TC, B => B_inst );
  end inst;

  -- pragma translate_off
configuration DW01_ash_inst_cfg_inst of DW01_ash_inst is
  for inst
  end for; -- inst
end DW01_ash_inst_cfg_inst;
-- pragma translate_on

```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_ash_inst( inst_A, inst_DATA_TC, inst_SH, inst_SH_TC, B_inst );

    parameter A_width = 8;
    parameter SH_width = 8;

    input [A_width-1 : 0] inst_A;
    input inst_DATA_TC;
    input [SH_width-1 : 0] inst_SH;
    input inst_SH_TC;
    output [A_width-1 : 0] B_inst;

    // Instance of DW01_ash
    DW01_ash #(A_width, SH_width)
        U1 (.A(inst_A), .DATA_TC(inst_DATA_TC), .SH(inst_SH),
           .SH_TC(inst_SH_TC), .B(B_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 6 ■ Added this Revision History table and the document links on this page

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